

Your Name and ID:

Your class: ☐ A (Prof. A. Pietracaprina)      ☐ B (Prof. F. Silvestri)

**REMINDERS:**

- Write your Name and ID on every sheet that you return
- Excessively long answers will be penalized
- TOTAL TIME: 2 hours

## PART 1

1. [**3 points**] Consider a MapReduce algorithm  $A$  that uses local space  $M_L$  and aggregate space  $M_A$ . In order to run  $A$  successfully on a cluster, what characteristics must the cluster have?
2. [**3 points**] Consider the execution of MR-Farthest-First Traversal to compute a solution to  $k$ -center for a set  $P$  of  $N$  points. Briefly describe Round 1 and Round 2, and analyze their local space requirements as a function of the number  $\ell$  of partitions.
3. [**4 points**] Consider a set  $P$  of  $n$  points from  $\mathbb{R}^D$  stored in a hash table  $T$  based on a hash function  $h$  randomly selected from a  $(p_1, p_2, c, r)$ -locality sensitive family  $\mathcal{H}$ .
  - (a) Explain how the hash table  $T$  can be used for answering a  $(c, r)$ -Approximate Near Neighbor Search query.
  - (b) Show that the query requires  $O(Dnp_2)$  time in expectation.
4. [**3 points**] Answer the following questions regarding the streaming framework.
  - (a) What are the performance metrics used to evaluate a streaming algorithm?
  - (b) As seen in Homework 3, the Spark Stream provides the items in batches. Briefly explain how each batch is represented and how a standard streaming algorithm (which processes one item at the time) can be implemented using Spark Stream.

## PART 2

1. **[7 points]** Let  $P$  be a set of  $N$  points in a metric space partitioned into  $k$  clusters  $C_1, C_2, \dots, C_k$ , with respective centers  $c_1, c_2, \dots, c_k$ . Each point  $x \in P$  is initially represented by the key-value pair  $(\text{ID}_x, (x, i_x))$ , where  $\text{ID}_x$  is a distinct key in  $[0, N-1]$ , and  $i_x$  is the index of the cluster of  $x$ . For a cluster  $C_i$ , its *furthest center* is the center  $c \in \{c_1, c_2, \dots, c_k\}$  that *maximizes* the average distance

$$\frac{1}{|C_i|} \sum_{x \in C_i} d(x, c).$$

- (a) Design an efficient *2-round MapReduce algorithm* that given  $P$  computes the  $k$  pairs  $(i, c)$ , where  $1 \leq i \leq k$  and  $c$  is the furthest center for  $C_i$ . Assume that  $k$  is a constant and that the centers  $c_1, c_2, \dots, c_k$  are global variables.
- (b) Analyze the local and aggregate space of your algorithm. (*For full score, local space must be  $o(N)$  and aggregate space  $O(N)$ .*)
2. **[6 points]** Let  $\Sigma$  be a stream of colored items  $(u_i, \gamma_i)$ , with  $1 \leq i \leq n$ , where  $u_i$  is an item from a universe  $U$ , and  $\gamma_i$  is its color (red or blue). We call  $f_{u, \text{red}}$  the frequency of the pair  $(u, \text{red})$  in  $\Sigma$ , and  $f_{u, \text{blue}}$  the frequency of the pair  $(u, \text{blue})$  in  $\Sigma$ . Also, for every  $u \in U$  define its *weighted frequency*  $f'_u$  as

$$f'_u = \frac{f_{u, \text{red}}}{2} + \frac{f_{u, \text{blue}}}{3}.$$

- (a) Describe how to obtain **one**  $d \times w$  array  $C$  of counters which can be employed to provide, for any  $u \in U$ , an unbiased estimate  $\tilde{f}'_u$  of its weighted frequency  $f'_u$ .
- (b) Consider an item  $u \in U$  and suppose that in row  $j$  of  $C$ , no other item uses the same cell as  $u$ . Does this row provide the exact  $f'_u$ ? Motivate your answer.